BUSINESS REQUIREMENTS DOCUMENT (BRD)

AI-Based Monitoring Solution for Oil & Gas Industry

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EXECUTIVE SUMMARY

This document outlines the business requirements for an AI-powered monitoring solution for Oil & Gas operations, integrating Machine Learning (ML), Deep Learning (DL), Computer Vision (CV), and Retrieval-Augmented Generation (RAG) technologies. The solution aims to reduce unplanned downtime by 50%, improve safety compliance by 40%, and enable predictive maintenance across upstream, midstream, and downstream operations.

PART 1: PROBLEM STATEMENT

1.1 Current State Analysis

1.1.1 Operational Challenges

Challenge	Description	Current Impact
Unplanned Downtime	Equipment failures without warning	$50M/year in lost production
Manual Inspections	Technicians physically inspect 5,000+ assets	2,000 man-hours/month
Safety Incidents	PPE violations, unsafe behaviors undetected	15 recordable incidents/year
Corrosion Detection	Visual inspection misses 30% of corrosion	$10M/year in pipeline repairs
Gas Leak Detection	Delayed detection of hydrocarbon leaks	3 major leak events/year
Documentation Access	Engineers spend 4 hours/day searching docs	40% productivity loss
Alarm Fatigue	10,000+ alarms/day, 95% nuisance	Critical alarms missed
Knowledge Loss	Retiring workforce (avg age 55+)	Expertise not captured

1.1.2 Financial Impact

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          ANNUAL COST OF CURRENT STATE

Unplanned Downtime (Production Loss)        $50,000,000
Emergency Maintenance & Repairs             $25,000,000
Safety Incidents (Fines, Medical, Legal)    $15,000,000
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Manual Inspection Labor                    $8,000,000
Corrosion-Related Failures                 $10,000,000
Inefficient Documentation Search           $5,000,000
Environmental Penalties (Leaks)            $12,000,000

TOTAL ANNUAL IMPACT                        $125,000,000
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1.1.3 Root Cause Analysis (5 Whys)

Problem: Critical equipment failure causing production shutdown

1. **Why?** Bearing failure in compressor
2. **Why?** Bearing degradation not detected
3. **Why?** Vibration data not analyzed in real-time
4. **Why?** No predictive analytics system in place
5. **Why?** Legacy systems cannot support AI/ML → **ROOT CAUSE**

1.2 Proposed Solution

1.2.1 Solution Overview

Integrated AI Monitoring Platform combining:

Technology	Application	Capability
Machine Learning (ML)	Predictive Maintenance	Failure prediction 30 days ahead
Deep Learning (DL)	Anomaly Detection	Real-time sensor anomaly detection
Computer Vision (CV)	Visual Inspection	Automated corrosion, PPE, leak detection
RAG (LLM + Retrieval)	Knowledge Assistant	Natural language access to 50,000+ docs

1.2.2 Solution Architecture Overview

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                    OIL & GAS AI MONITORING PLATFORM


   SENSORS          CAMERAS          SCADA          DOCUMENTS
Vibration          PTZ 4K          Historian        Manuals
Temperature        Thermal         OPC-UA           P&IDs
Pressure           Drone           Modbus           SOPs
Flow               Fixed                            Reports



                    EDGE COMPUTING LAYER
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- Real-time inference (NVIDIA Jetson)
- Data preprocessing & filtering
- Local anomaly detection
- Video analytics (CV models)

CLOUD PLATFORM LAYER

ML	DL	CV	RAG
Predictive Maintenance	Anomaly Detection	Visual Inspection	Knowledge Assistant

DATA LAKE & ANALYTICS ENGINE

- Time-series: InfluxDB (sensor data)
- Vector DB: Milvus (embeddings for RAG)
- Object Storage: S3 (images, videos, models)
- Graph DB: Neo4j (asset relationships)

APPLICATION LAYER

- Control Room Dashboard (Real-time monitoring)
- Mobile App (Field technicians)
- RAG Chatbot (Document queries)
- Alert Management (SMS, Email, Push)
- Integration (SAP PM, Maximo, OSIsoft PI)

1.3 Success Criteria (KPIs)

1.3.1 Primary KPIs

KPI	Baseline	Target	Measurement Method
Unplanned Downtime	500 hours/year	250 hours/year (-50%)	CMMS tracking
Failure Prediction Accuracy	N/A	>90%	Confusion matrix
Mean Time to Detection (MTTD)	8 hours	15 minutes	Alert timestamp

KPI	Baseline	Target	Measurement Method
Safety Incident Rate (TRIR)	2.5	1.5 (-40%)	HSE records
PPE Compliance	75%	98%	CV detection logs
Corrosion Detection Rate	70%	95%	POD validation
Document Search Time	4 hours/query	30 seconds/query	RAG logs
Alarm Reduction (Actionable)	5% actionable	60% actionable	Alarm analytics

1.3.2 Secondary KPIs

KPI	Target	Measurement
Model Inference Latency	<100ms	Prometheus metrics
System Availability	99.9%	Uptime monitoring
User Adoption Rate	>80%	Usage analytics
ROI	300% in Year 3	Financial analysis

1.4 Stakeholders

1.4.1 Stakeholder Matrix

Stakeholder	Role	Interest	Influence	Engagement
VP Operations	Executive Sponsor	High	High	Monthly steering
Plant Manager	Business Owner	High	High	Weekly updates
Reliability Engineer	Primary User	High	Medium	Daily use
Control Room Operator	Primary User	High	Medium	Daily use
Field Technician	Mobile User	Medium	Low	Daily use
HSE Manager	Safety Oversight	High	Medium	Weekly reviews
IT Director	Technical Owner	Medium	High	Architecture decisions
Data Scientist	Model Development	High	Medium	Daily development

Stakeholder	Role	Interest	Influence	Engagement
Maintenance Planner	Work Order Integration	Medium	Low	SAP integration

1.4.2 RACI Matrix

Activity	VP Ops	Plant Mgr	IT Dir	Data Sci	Reliability Eng
Budget Approval	A	R	C	I	I
Architecture Design	I	C	A	R	C
Model Development	I	I	C	R/A	C
User Acceptance	I	A	I	C	R
Go-Live Decision	A	R	C	I	C

R=Responsible, A=Accountable, C=Consulted, I=Informed

1.5 Constraints

1.5.1 Budget Constraints

Category	Allocation
Hardware (Edge + Cloud)	$2,000,000
Software Licenses	$500,000/year
Professional Services	$1,500,000
Internal Resources	$1,000,000
Training	$200,000
Contingency (15%)	$780,000
TOTAL	**$5,980,000**

1.5.2 Timeline Constraints

- **Phase 1 (Pilot)**: 6 months - Single facility
- **Phase 2 (Rollout)**: 12 months - All facilities
- **Go-Live Deadline**: December 2026 (Regulatory compliance)

1.5.3 Technical Constraints

Constraint	Implication	Mitigation
ATEX Zone 1 areas	Explosion-proof hardware required	IS barriers, ATEX-certified cameras
Limited bandwidth (offshore)	Edge-heavy processing	Local inference, compressed uploads
Legacy DCS (1990s)	No native connectivity	OPC-DA to OPC-UA gateway
No public cloud allowed	On-premise deployment	Private cloud, VMware
Data sovereignty	Data must stay in-country	Local data centers

1.5.4 Regulatory Constraints

Regulation	Requirement	Impact
API 580/581 RBI	Risk-based inspection	CV models must meet POD standards
OSHA 1910.119 PSM	Process safety management	ML models need MOC approval
EPA LDAR	Leak detection reporting	Automated leak reports
GDPR (if applicable)	Personnel data protection	Face blur in CV

1.6 Risks and Mitigations

1.6.1 Risk Register

ID	Risk	Probability	Impact	Score	Mitigation	Owner
R1	Model accuracy below target	Medium	High	12	Phased deployment, extensive testing	Data Scientist
R2	Edge hardware failure in harsh environment	Medium	Medium	9	Redundant hardware, MTBF >100K hours	Hardware Engineer
R3	User adoption resistance	High	Medium	12	Change management, training	Plant Manager
R4	Integration with legacy SCADA fails	Medium	High	12	OPC gateway, protocol adapters	IT Director
R5	Cybersecurity breach	Low	Critical	12	Air-gapped network, IEC 62443	Security Engineer
R6	Regulatory non-compliance	Low	Critical	12	Early HSE engagement, MOC process	HSE Manager
R7	Vendor lock-in	Medium	Medium	9	Open-source stack, multi-vendor	Solution Architect

ID	Risk	Probability	Impact	Score	Mitigation	Owner
R8	Scope creep	High	Medium	12	Strict change control, MVP focus	Project Manager

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1.7 Acceptance Criteria

1.7.1 Functional Requirements

ID	Requirement	Acceptance Criteria
FR-01	Predictive maintenance for rotating equipment	Predict failures >7 days ahead with 90% accuracy
FR-02	Real-time anomaly detection	Detect anomalies within 60 seconds, <1% false positive
FR-03	Corrosion detection from images	Detect corrosion >0.5mm with 95% POD
FR-04	PPE compliance monitoring	Detect 5 PPE types with 98% accuracy
FR-05	Gas leak detection	Detect leaks >100 ppm within 30 seconds
FR-06	Natural language document query	Return relevant results in <5 seconds, 90% relevance
FR-07	Alarm rationalization	Reduce nuisance alarms by 80%
FR-08	Mobile app for technicians	Offline capability, sync when connected
FR-09	SAP PM integration	Automatic work order creation
FR-10	Dashboard visualization	Real-time update <5 second latency

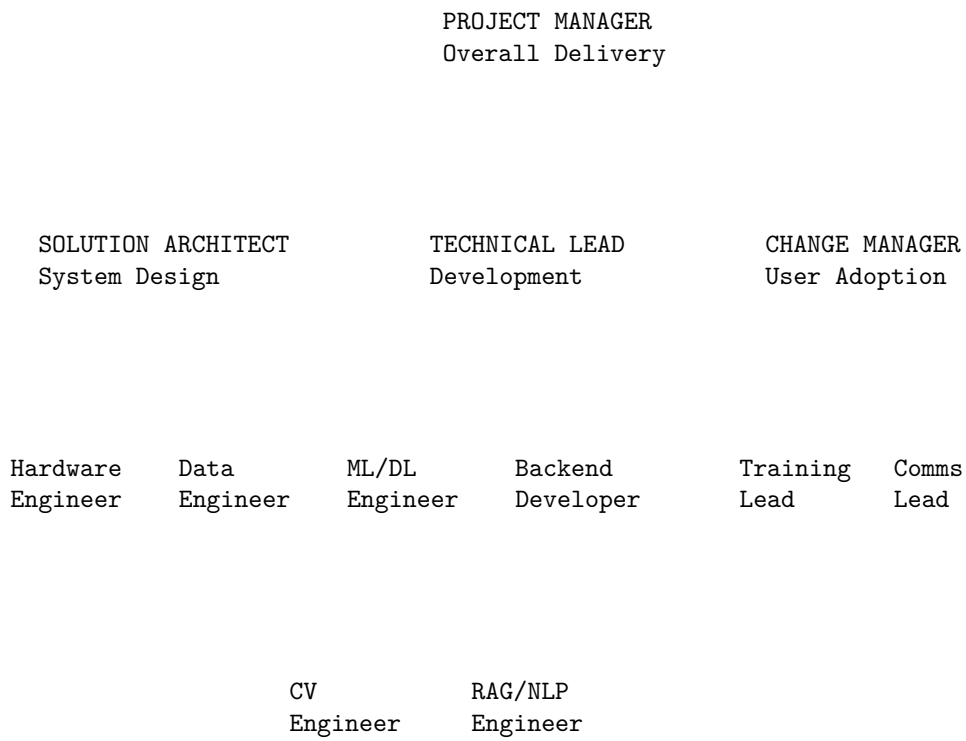
1.7.2 Non-Functional Requirements

ID	Requirement	Acceptance Criteria
NFR-01	System availability	99.9% uptime (8.76 hours downtime/year max)
NFR-02	Inference latency	<100ms edge, <500ms cloud
NFR-03	Data retention	7 years raw data, 10 years analytics
NFR-04	Scalability	Support 10x data growth
NFR-05	Security	IEC 62443 SL2 compliance
NFR-06	Disaster recovery	RPO: 1 hour, RTO: 4 hours

PART 2: ROLES AND RESPONSIBILITIES

2.1 Project Team Structure

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STEERING COMMITTEE
VP Operations
IT Director
HSE Director
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2.2 Role Descriptions

2.2.1 Project Manager

Reports to: Steering Committee

Responsibility	Deliverables
Project planning and tracking	Project plan, WBS, Gantt chart
Budget management	Monthly budget report
Risk management	Risk register, mitigation actions
Stakeholder communication	Weekly status report, steering deck
Vendor management	Contract management, SLAs
Change control	Change request log

Tools: Jira, MS Project, Confluence, Power BI

2.2.2 Solution Architect

Reports to: Project Manager

Responsibility	Deliverables
System architecture design	Architecture document, diagrams
Technology selection	Technology evaluation matrix
Integration design	Integration specifications
Security architecture	Security design document
Performance design	Capacity planning document

Tools: Lucidchart, ArchiMate, Sparx EA

2.2.3 Hardware Engineer

Reports to: Solution Architect

Responsibility	Deliverables
Sensor selection	Sensor specification sheet
Edge hardware design	Edge hardware BOM
Network design	Network topology diagram
ATEX compliance	Hazardous area classification
Installation supervision	Installation checklist
Commissioning	Commissioning report

Tools: AutoCAD, KiCad, Wireshark

2.2.4 Data Engineer

Reports to: Technical Lead

Responsibility	Deliverables
Data pipeline design	Data flow diagrams
ETL development	ETL jobs, Airflow DAGs
Data lake architecture	Data lake schema
Data quality	Data quality reports
Historian integration	PI/Historian connectors

Tools: Apache Kafka, Spark, Airflow, dbt

2.2.5 ML/DL Engineer

Reports to: Technical Lead

Responsibility	Deliverables
ML model development	Trained models, notebooks
Model optimization	Quantized models, benchmarks

Responsibility	Deliverables
Model deployment	MLflow registry, APIs
Model monitoring	Drift detection dashboards
A/B testing	Experiment reports

Tools: PyTorch, TensorFlow, MLflow, Weights & Biases

2.2.6 CV Engineer

Reports to: ML/DL Engineer

Responsibility	Deliverables
Image dataset creation	Labeled datasets
Object detection models	YOLOv8 trained models
Video analytics pipeline	GStreamer pipelines
Edge deployment	TensorRT optimized models
Accuracy validation	POD curves, confusion matrices

Tools: CVAT, Label Studio, OpenCV, DeepStream

2.2.7 RAG/NLP Engineer

Reports to: ML/DL Engineer

Responsibility	Deliverables
Document ingestion pipeline	Parsing scripts, loaders
Embedding model selection	Embedding benchmarks
Vector database setup	Milvus/Pinecone deployment
LLM fine-tuning	LoRA adapters
RAG pipeline development	LangChain implementation
Evaluation	RAGAS scores, user feedback

Tools: LangChain, LlamaIndex, Milvus, Hugging Face

2.2.8 Backend Developer

Reports to: Technical Lead

Responsibility	Deliverables
API development	FastAPI endpoints
Database design	Schema, migrations
Authentication/Authorization	Keycloak integration
Integration development	SAP, Maximo connectors

Responsibility	Deliverables
Microservices	Docker containers

Tools: FastAPI, PostgreSQL, Redis, Docker

2.2.9 Frontend Developer

Reports to: Technical Lead

Responsibility	Deliverables
Dashboard development	React web app
Mobile app development	React Native app
Data visualization	Grafana dashboards
UX design	Wireframes, prototypes

Tools: React, React Native, Grafana, Figma

2.2.10 DevOps Engineer

Reports to: Technical Lead

Responsibility	Deliverables
CI/CD pipeline	GitHub Actions, Jenkins
Infrastructure as Code	Terraform, Ansible
Kubernetes deployment	Helm charts
Monitoring setup	Prometheus, Grafana
Security hardening	Vulnerability scans

Tools: Kubernetes, Terraform, ArgoCD, Prometheus

2.2.11 QA Engineer

Reports to: Technical Lead

Responsibility	Deliverables
Test strategy	Test plan document
Automated testing	Pytest, Selenium scripts
Performance testing	JMeter reports
UAT coordination	UAT scripts, sign-off

Tools: Pytest, Selenium, JMeter, Postman

2.2.12 HSE Specialist

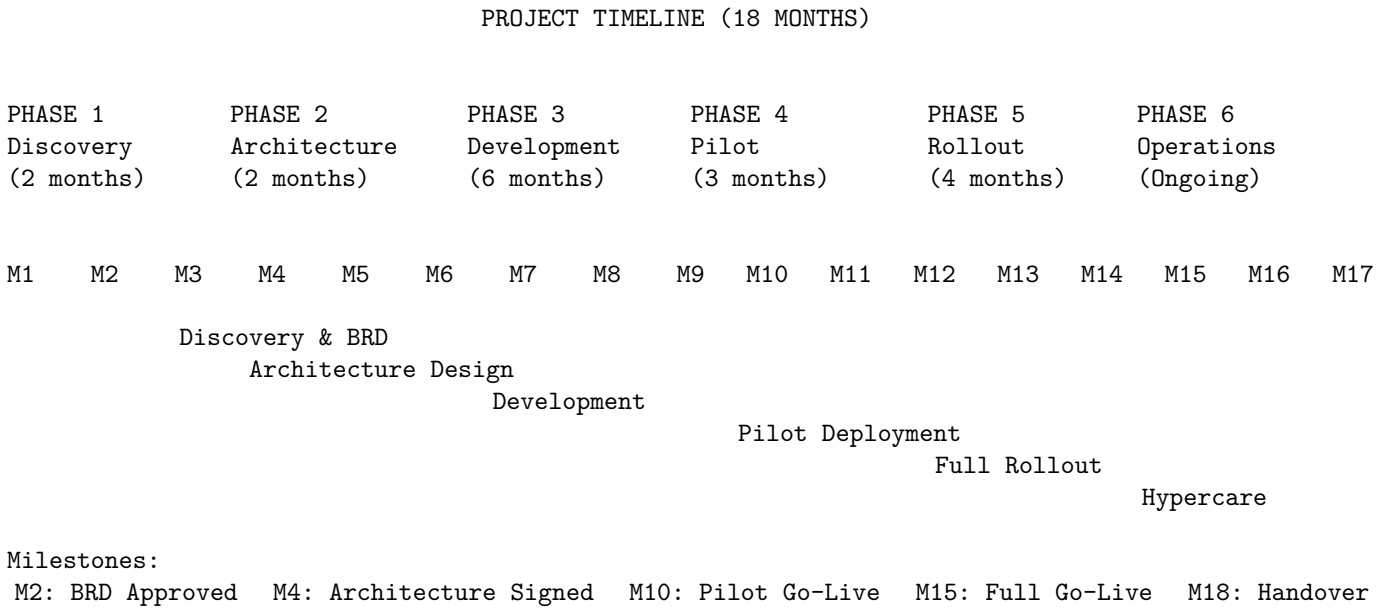
Reports to: Project Manager

Responsibility	Deliverables
Safety requirements	HSE requirements doc
MOC process	Management of Change forms
HAZOP review	HAZOP reports
Compliance verification	Compliance checklist

Tools: BowTie, PHAWorks

PART 3: IMPLEMENTATION PHASES

Phase Overview



PHASE 1: DISCOVERY & REQUIREMENTS (Months 1-2)

1.1 Objectives

- Understand current operations and pain points
- Document detailed requirements
- Finalize scope and success criteria
- Approve BRD and project charter

1.2 Activities & Tasks

Week	Activity	Tasks	Responsible	Deliverable
W1-2	Stakeholder Workshops	Conduct interviews with 15+ stakeholders	PM, BA	Interview notes
W1-2	Current State Analysis	Document existing systems, data flows	Data Engineer	As-is architecture
W2-3	Requirements Gathering	Functional/Non-functional requirements	BA, Solution Architect	Requirements doc
W3-4	Data Audit	Assess data availability, quality, gaps	Data Engineer	Data audit report
W4-5	Use Case Prioritization	MoSCoW prioritization workshop	PM, Stakeholders	Prioritized use case list
W5-6	Success Metrics	Define KPIs, baselines, targets	PM, Plant Manager	KPI dashboard design
W6-7	Risk Assessment	Identify risks, mitigations	PM, HSE	Risk register
W7-8	BRD Finalization	Document review, approval	All	Approved BRD

1.3 Deliverables

- ☒ Stakeholder interview summaries
- ☒ Current state architecture diagram
- ☒ Requirements specification document
- ☒ Data audit report
- ☒ Prioritized use case list
- ☒ Risk register
- ☒ Approved BRD

1.4 Gate Criteria

Criteria	Pass/Fail
BRD approved by Steering Committee	Required
Budget confirmed	Required
Project team confirmed	Required
Data availability confirmed for MVP	Required

PHASE 2: ARCHITECTURE & DESIGN (Months 3-4)

2.1 Objectives

- Design end-to-end system architecture
- Select technologies and vendors
- Create detailed design documents
- Plan pilot deployment

2.2 Activities & Tasks

Week	Activity	Tasks	Responsible	Deliverable
W1-2	Architecture Design	Design edge, cloud, integration layers	Solution Architect	Architecture document
W1-2	Technology Evaluation	Evaluate ML platforms, databases, edge HW	Technical Lead	Technology matrix
W2-3	Vendor Selection	RFP for hardware, cloud, professional services	PM, Procurement	Vendor contracts
W3-4	Security Design	IEC 62443 compliance, network segmentation	Security Engineer	Security architecture
W3-4	Data Architecture	Data lake design, schema, retention	Data Engineer	Data architecture doc
W4-5	Integration Design	SAP PM, Historian, SCADA interfaces	Backend Developer	Integration specs
W5-6	ML Architecture	Model serving, MLOps, monitoring	ML Engineer	ML architecture doc
W6-7	Hardware BOM	Finalize edge hardware, sensors, network	Hardware Engineer	BOM, procurement list

Week	Activity	Tasks	Responsible	Deliverable
W7-8	Design Review	Architecture review with stakeholders	All	Approved architecture

2.3 Technology Selection

2.3.1 Edge Computing

Component	Selected	Alternatives	Rationale
Edge Computer	NVIDIA Jetson AGX Orin 64GB	Advantech, Dell Edge	Best ML performance/watt
Edge Gateway	Cisco IE4000	Siemens, Hirschmann	Industrial-grade, OPC-UA
Vibration DAQ	NI cDAQ-9185 + NI-9234	B&K, IMI	High-freq, industrial

2.3.2 Cloud/On-Premise

Component	Selected	Alternatives	Rationale
ML Platform	Kubeflow on Kubernetes	AWS SageMaker, Azure ML	On-premise requirement
Time-Series DB	InfluxDB Enterprise	TimescaleDB, OSIsoft PI	Native ML integration
Vector DB	Milvus	Pinecone, Weaviate	On-premise, open-source
LLM	Llama 3.1 70B	GPT-4, Claude	On-premise, fine-tunable

2.3.3 ML/AI Frameworks

Capability	Framework	Version
ML Training	PyTorch	2.1
CV Inference	TensorRT	8.6
Video Analytics	NVIDIA DeepStream	6.3
RAG	LangChain	0.1.x
MLOps	MLflow	2.x

2.4 Deliverables

- ☒ Solution architecture document
- ☒ Technology selection matrix
- ☒ Vendor contracts
- ☒ Security architecture
- ☒ Data architecture
- ☒ Integration specifications
- ☒ Hardware BOM
- ☒ Network design

2.5 Gate Criteria

Criteria	Pass/Fail
Architecture approved	Required
Vendors contracted	Required
Hardware ordered	Required
Security review passed	Required

PHASE 3: DEVELOPMENT (Months 5-10)

3.1 Objectives

- Develop all AI/ML models
- Build data pipelines and integrations
- Develop frontend applications
- Prepare for pilot deployment

3.2 Sprint Planning (Agile - 2-week sprints)

Sprint	Focus Area	Key Deliverables
S1-S2	Data Pipeline	Kafka setup, SCADA integration, data ingestion
S3-S4	ML: Predictive Maintenance	Vibration model training, validation
S5-S6	DL: Anomaly Detection	Autoencoder training, edge deployment
S7-S8	CV: Corrosion Detection	Dataset labeling, YOLO training
S9-S10	CV: PPE Detection	PPE model training, video pipeline
S11-S12	RAG: Document Ingestion	PDF parsing, embedding, vector DB
S13-S14	RAG: Query Interface	LLM integration, chatbot UI
S15-S16	Dashboard & Mobile	Control room UI, mobile app
S17-S18	Integration	SAP PM, alarm management
S19-S20	Testing & Hardening	End-to-end testing, security audit

3.3 Detailed Development Tasks

3.3.1 Data Pipeline Development

Task	Description	Tools	Duration
Kafka Setup	Deploy Kafka cluster, configure topics	Apache Kafka	1 week
SCADA Integration	OPC-UA to Kafka connector	Apache NiFi, OPC-UA	2 weeks
Historian Bridge	OSIsoft PI to InfluxDB sync	PI AF SDK, Python	2 weeks
Data Validation	Schema validation, quality checks	Great Expectations	1 week
Feature Store	Time-series feature engineering	Feast, Spark	2 weeks

3.3.2 ML Model Development — Predictive Maintenance Model

Step	Task	Duration
1	Data collection (6 months historical vibration)	2 weeks
2	Feature engineering (RMS, kurtosis, FFT)	2 weeks
3	Model training (XGBoost, LSTM, Transformer)	3 weeks
4	Hyperparameter optimization (Optuna)	1 week
5	Model validation (cross-validation, backtesting)	2 weeks
6	Quantization & TensorRT optimization	1 week
7	Edge deployment & testing	1 week

Anomaly Detection Model

Step	Task	Duration
1	Baseline data collection (normal operation)	2 weeks
2	Autoencoder architecture design	1 week
3	Training (reconstruction loss)	2 weeks
4	Threshold calibration	1 week
5	Real-time inference pipeline	1 week

3.3.3 Computer Vision Development — Corrosion Detection

Step	Task	Duration
1	Image collection (drone, fixed cameras)	3 weeks
2	Annotation (CVAT - 5000 images)	3 weeks
3	YOLOv8 training	2 weeks
4	Data augmentation & retraining	1 week
5	POD validation	1 week
6	Edge deployment (Jetson, TensorRT)	1 week

PPE Detection

Step	Task	Duration
1	Video data collection	2 weeks
2	Annotation (helmet, glasses, gloves, vest, boots)	2 weeks
3	YOLOv8 training	2 weeks
4	Tracking integration (ByteTrack)	1 week
5	Privacy masking (face blur)	1 week

Gas Leak Detection

Step	Task	Duration
1	Thermal + RGB camera setup	1 week
2	Synthetic leak data generation	2 weeks
3	Multi-modal fusion model	2 weeks
4	Alert logic & integration	1 week

3.3.4 RAG Development

Step	Task	Duration
1	Document collection (50,000 docs)	2 weeks
2	PDF/CAD parsing (Unstructured, PyMuPDF)	2 weeks
3	Chunking strategy design	1 week
4	Embedding generation (BGE-large)	1 week
5	Milvus vector DB setup	1 week
6	LLM fine-tuning (Llama 3.1 on O&G data)	2 weeks
7	RAG pipeline (LangChain)	2 weeks
8	Chatbot UI development	2 weeks
9	Evaluation (RAGAS metrics)	1 week

3.4 Deliverables

- ☒ Data ingestion pipelines
- ☒ Trained ML models (predictive maintenance)
- ☒ Trained DL models (anomaly detection)
- ☒ Trained CV models (corrosion, PPE, leak)
- ☒ RAG system with chatbot
- ☒ Control room dashboard
- ☒ Mobile application
- ☒ SAP PM integration
- ☒ Test reports

3.5 Gate Criteria

Criteria	Pass/Fail
All models meet accuracy targets	Required
Security audit passed	Required
Integration tests passed	Required
Performance tests passed	Required

PHASE 4: PILOT DEPLOYMENT (Months 11-13)

4.1 Objectives

- Deploy to single facility
- Validate in production environment
- Gather user feedback
- Refine models and processes

4.2 Pilot Scope

Item	Pilot Scope
Facility	Refinery Unit A (100 assets)
Duration	3 months

Item	Pilot Scope
Users	10 operators, 5 engineers, 2 managers
Models	All 4 AI capabilities
Success Criteria	See KPI targets

4.3 Pilot Activities

Week	Activity	Tasks
W1-2	Hardware Installation	Edge devices, cameras, sensors
W2-3	Network Configuration	VLANs, firewalls, OPC connectivity
W3-4	Model Deployment	Deploy models to edge and cloud
W4-5	Integration Testing	End-to-end data flow validation
W5-6	User Training	Control room, field technicians
W6-8	Shadow Mode	Run in parallel, no action taken
W8-10	Live Mode	Alerts trigger actions
W10-12	Feedback & Refinement	Model tuning based on feedback

4.4 Pilot Metrics Tracking

Metric	Week 4	Week 8	Week 12	Target
Failure predictions	2	5	10	>5
True positives (predictive)	50%	70%	90%	>90%
False positives (anomaly)	15%	8%	2%	<2%
PPE compliance	80%	90%	98%	>98%
RAG query accuracy	70%	85%	92%	>90%
User satisfaction (1-5)	3.0	3.5	4.2	>4.0

4.5 Deliverables

- ☒ Installed and configured pilot system
- ☒ Trained users
- ☒ Pilot results report
- ☒ Refined models
- ☒ Go/No-Go recommendation

4.6 Gate Criteria

Criteria	Pass/Fail
All KPIs met or exceeded	Required
No safety incidents	Required
User adoption >70%	Required
Steering approval for rollout	Required

PHASE 5: FULL ROLLOUT (Months 14-17)

5.1 Objectives

- Deploy to all facilities
- Scale infrastructure
- Complete training rollout
- Establish support model

5.2 Rollout Plan

Month	Facilities	Assets	Users
M14	Refinery B, C	200	50
M15	Offshore A, B	150	40
M16	Pipeline Network	300	30
M17	Gas Processing	250	40
Total	**All 7 facilities**	**1,000**	**170**

5.3 Rollout Activities

Activity	Description	Duration
Site Survey	Validate infrastructure, network	1 week/site
Hardware Deployment	Edge devices, cameras	2 weeks/site
Model Customization	Retrain on site-specific data	2 weeks/site
Integration	SCADA, Historian, SAP	1 week/site
Training	Site-specific training	3 days/site
Go-Live	Monitored go-live	1 week/site

5.4 Deliverables

☒ All facilities live
☒ All users trained
☒ Support runbook
☒ SLA documentation

PHASE 6: OPERATIONS & CONTINUOUS IMPROVEMENT (Month 18+)

6.1 Support Model

Level	Response Time	Responsibility
L1	<1 hour	Internal IT helpdesk
L2	<4 hours	Internal platform team
L3	<24 hours	Vendor/Development team

6.2 Ongoing Activities

Activity	Frequency	Responsible
Model retraining	Monthly	ML Engineer
Model drift monitoring	Continuous	MLOps
User feedback review	Weekly	Product Owner
KPI reporting	Weekly	PM
Security patching	Monthly	DevOps
New use case development	Quarterly	Solution Architect

6.3 Continuous Improvement

- Monthly model performance reviews
- Quarterly user satisfaction surveys
- Annual roadmap planning
- New AI capability evaluation (GPT-5, etc.)

PART 4: TOOLS AND TECHNOLOGIES

4.1 Complete Technology Stack

4.1.1 Hardware

Category	Component	Model	Qty	Unit Cost	Total
Edge Compute	AI Computer	Jetson AGX Orin 64GB	20	$2,199	$43,980
Edge Gateway	Industrial Gateway	Cisco IE4000	20	$3,500	$70,000
Vibration Sensor	Accelerometer	PCB 352C33	200	$600	$120,000
Temperature	RTD	Omega PT100	200	$50	$10,000
Pressure	Transmitter	Rosemount 3051	100	$1,500	$150,000
Camera (Visual)	Industrial 4K	Axis P1448-LE	50	$1,200	$60,000
Camera (Thermal)	Thermal	FLIR A700	20	$8,000	$160,000
DAQ	High-speed	NI cDAQ-9185 + NI-9234	20	$4,000	$80,000
Network	Industrial Switch	Cisco IE3400	30	$2,000	$60,000
Server	GPU Server	Dell R750xa + A100	4	$50,000	$200,000
Storage	NAS	NetApp AFF A250	2	$80,000	$160,000
TOTAL HARDWARE					**$1,113,980**

4.1.2 Software

Category	Tool	License	Annual Cost
Container Orchestration	Kubernetes (OpenShift)	Enterprise	$100,000
ML Platform	Kubeflow	Open Source	$0
Time-Series DB	InfluxDB Enterprise	Enterprise	$50,000
Vector DB	Milvus	Open Source	$0
Monitoring	Prometheus + Grafana	Open Source	$0
LLM	Llama 3.1 70B	Open Source	$0
MLOps	MLflow	Open Source	$0
ETL	Apache NiFi	Open Source	$0
Streaming	Apache Kafka	Confluent	$80,000
Dashboard	Grafana Enterprise	Enterprise	$20,000
Mobile	React Native	Open Source	$0
TOTAL SOFTWARE			**$250,000/year**

4.1.3 Development Tools

Purpose	Tool	License
IDE	VS Code, PyCharm	Free/Commercial
Version Control	GitHub Enterprise	Commercial
CI/CD	GitHub Actions, ArgoCD	Free
Documentation	Confluence	Commercial
Project Management	Jira	Commercial
Design	Figma, Lucidchart	Commercial
Annotation	CVAT, Label Studio	Open Source
Notebooks	JupyterHub	Open Source
Testing	Pytest, Selenium	Open Source

4.2 Tool-Role Mapping

Role	Primary Tools	Secondary Tools
Project Manager	Jira, Confluence, MS Project	Power BI, Teams
Solution Architect	Lucidchart, ArchiMate	Confluence, Visio
Hardware Engineer	KiCad, AutoCAD	Oscilloscope, Logic Analyzer
Data Engineer	Kafka, Spark, Airflow	dbt, Great Expectations
ML Engineer	PyTorch, MLflow, W&B	Optuna, TensorRT
CV Engineer	CVAT, DeepStream, OpenCV	Label Studio, Roboflow
RAG Engineer	LangChain, Milvus, HuggingFace	ChromaDB, LlamaIndex
Backend Developer	FastAPI, PostgreSQL, Redis	Docker, Postman
Frontend Developer	React, Grafana, React Native	Figma, Storybook
DevOps Engineer	Kubernetes, Terraform, ArgoCD	Prometheus, Vault

Role	Primary Tools	Secondary Tools
QA Engineer	Pytest, Selenium, JMeter	Postman, SonarQube

PART 5: CHALLENGES, SOLUTIONS, AND STRATEGIES

5.1 Technical Challenges

Challenge 1: Data Quality and Availability

Aspect	Challenge	Impact	Solution	Strategy
Missing Data	30% of sensors have gaps	Model accuracy degraded	Imputation algorithms	Use KNN imputation, flag low-confidence predictions
Noisy Data	EMI in plant causes spikes	False positives	Signal filtering	Butterworth filter, outlier detection (IQR)
Label Scarcity	Few labeled failure events	Supervised training difficult	Semi-supervised learning	Use autoencoders, synthetic data generation
Data Silos	Data in separate systems	Incomplete picture	Unified data lake	Kafka + data lake architecture

Challenge 2: Edge Deployment in Hazardous Areas

Aspect	Challenge	Impact	Solution	Strategy
ATEX Compliance	Zone 1 requires Ex-rated	Standard hardware not allowed	IS barriers, Ex-rated enclosures	Partner with Ex-certified vendor
Harsh Environment	Temp, dust, vibration	Hardware failure	Industrial-grade IP67 hardware	MTBF >100,000 hours requirement
Limited Power	Remote locations	No reliable power	Solar + battery	Design for 10W power budget
Connectivity	No wired network	Data transmission	LoRaWAN, Satellite	Edge-heavy processing, compressed uploads

Challenge 3: Model Accuracy and Generalization

Aspect	Challenge	Impact	Solution	Strategy
Domain Shift	Models trained on one plant fail on another	Poor accuracy on new sites	Transfer learning	Fine-tune on site-specific data
Rare Events	Failures are infrequent	Imbalanced datasets	Data augmentation, SMOTE	Oversample rare classes, synthetic generation
Concept Drift	Equipment behavior changes over time	Model degradation	Continuous monitoring	Automated retraining pipeline
Explainability	Black-box models	User distrust	XAI integration	SHAP values, attention visualization

Challenge 4: Integration with Legacy Systems

Aspect	Challenge	Impact	Solution	Strategy
Old Protocols	Modbus, OPC-DA (1990s)	No direct connectivity	Protocol converters	Deploy OPC-DA to OPC-UA gateways
Closed Systems	Proprietary historian	Limited API access	Vendor engagement	Negotiate API access or use ODBC
Data Formats	Inconsistent naming, units	Data quality issues	Standardization layer	Develop semantic mapping layer
Change Resistance	IT/OT convergence friction	Project delays	Stakeholder management	Joint IT/OT governance

5.2 Organizational Challenges

Challenge 5: User Adoption

Aspect	Challenge	Impact	Solution	Strategy
Change Resistance	“We’ve always done it this way”	Low usage	Change management	Executive sponsorship, quick wins
Skill Gap	Users unfamiliar with AI	Misinterpretation	Training programs	Role-based training, ongoing support
Trust Issues	Distrust of AI recommendations	Override AI	Transparency	Show confidence scores, explain reasoning
Workload Concerns	Fear of job displacement	Resistance	Communication	Position as augmentation, not replacement

Challenge 6: Regulatory Compliance

Aspect	Challenge	Impact	Solution	Strategy
MOC Process	AI changes require formal review	Deployment delays	Early engagement	Include HSE in design, pre-approve model updates
Audit Trail	Need to explain AI decisions	Compliance risk	Logging infrastructure	Log all predictions, inputs, outputs
Validation	Models need validation like instruments	Expensive process	Standardized validation	Develop validation protocol, automate testing

5.3 Strategic Approaches

Strategy 1: Phased Deployment with Quick Wins

```
Phase 1: PPE Detection (Visible, easy to validate)
    ↓ Success builds confidence
Phase 2: Anomaly Detection (Value demonstrated)
    ↓ Trust established
Phase 3: Predictive Maintenance (High complexity, high value)
    ↓ Full adoption
Phase 4: RAG Knowledge Assistant (Cultural change)
```

Strategy 2: Human-in-the-Loop Design

```
AI System → Recommendation → Human Review → Action
    ↑                              ↓
        Feedback Loop
```

- AI suggests, human decides
- Confidence thresholds for escalation
- Feedback improves models
- Gradual autonomy increase

Strategy 3: Fail-Safe Architecture

```
                    AI MONITORING LAYER
  (Can fail without operational impact)

                    EXISTING CONTROL LAYER
  (DCS/SCADA continues to operate independently)

                    PHYSICAL SAFEGUARDS
  (Pressure relief, ESD - always functional)
```

- AI enhances but doesn't replace safety systems
- Graceful degradation on AI failure
- Manual override always available

PART 6: HARDWARE SELECTION GUIDE

6.1 Sensor Selection Matrix

6.1.1 Vibration Sensors

Parameter	Option A	Option B	Option C	Selected
Model	PCB 352C33	IMI 603C01	B&K 4507	**PCB 352C33**
Sensitivity	100 mV/g	100 mV/g	100 mV/g	✓
Frequency Range	0.5-10,000 Hz	0.5-10,000 Hz	0.3-6,000 Hz	Wider range
Temperature Range	-54 to 121°C	-54 to 121°C	-54 to 100°C	Wider range
ATEX Certified	Yes	Yes	Yes	✓
Price	$600	$450	$800	Best value
Decision				**Selected**

6.1.2 Cameras

Parameter	Visual	Thermal	Decision Criteria
Model	Axis P1448-LE	FLIR A700	
Resolution	4K (8MP)	640×480	Visual for detail
Frame Rate	30 fps	30 fps	Both adequate
Operating Temp	-40 to 50°C	-40 to 55°C	Both suitable
ATEX Option	Zone 2 housing	Zone 2 housing	Available
Analytics	ACAP support	FLIR API	Edge processing
Price	$1,200	$8,000	Thermal for leaks
Use Case	PPE, corrosion	Gas leaks	Complementary

6.1.3 Edge Compute

Parameter	Jetson AGX Orin 64GB	Advantech MIC-730AI	Intel NUC Edge AI	Selected
AI Performance	275 TOPS	32 TOPS	12 TOPS	**Best**
Power	60W	50W	25W	Acceptable
Memory	64GB	16GB	32GB	Best
Operating Temp	-25 to 50°C	-20 to 60°C	0 to 35°C	Industrial
ATEX Housing	Available	Available	Limited	Yes
Price	$2,199	$2,500	$1,200	Best value
Decision				**Selected**

6.2 Complete Bill of Materials (BOM)

6.2.1 Per-Asset Sensor Kit

Item	Model	Qty	Unit Cost	Extended
Vibration Sensor	PCB 352C33	2	$600	$1,200
Temperature Sensor	PT100 RTD	2	$50	$100
Signal Conditioner	4-20mA	4	$75	$300
Junction Box	IP66 SS	1	$150	$150
Cable (shielded)	100m	1	$200	$200
Per Asset Total				**$1,950**

6.2.2 Edge Node Kit

Item	Model	Qty	Unit Cost	Extended
Edge Computer	Jetson AGX Orin 64GB	1	$2,199	$2,199
Industrial Enclosure	ATEX Zone 2	1	$800	$800
Power Supply	24VDC 100W	1	$150	$150
DAQ Module	NI-9234	1	$1,500	$1,500
Network Switch	Industrial 8-port	1	$400	$400
UPS	600VA	1	$300	$300
Per Edge Node Total				**$5,349**

6.2.3 Camera Kit

Item	Model	Qty	Unit Cost	Extended
Visual Camera	Axis P1448-LE	1	$1,200	$1,200
Thermal Camera	FLIR A700	1	$8,000	$8,000
Housing (ATEX)	Axis XP40-Q	1	$2,500	$2,500
Mounting Bracket	SS Heavy Duty	2	$100	$200
PoE Injector	Industrial	2	$150	$300
Per Camera Location Total				**$12,200**

6.2.4 Total Hardware Cost Summary

Category	Units	Unit Cost	Total Cost
Sensor Kits	200 assets	$1,950	$390,000
Edge Nodes	20 locations	$5,349	$106,980
Camera Kits	30 locations	$12,200	$366,000
Central Servers	4 GPU servers	$50,000	$200,000
Storage	2 NAS systems	$80,000	$160,000
Networking	1 lot	$100,000	$100,000
TOTAL HARDWARE			**$1,322,980**
Contingency (15%)			$198,447
GRAND TOTAL			**$1,521,427**

PART 7: DATABASE SCHEMA FOR PROJECT TRACKING

7.1 Entity Relationship Diagram



7.2 SQL Schema

```sql
-- Projects Table
CREATE TABLE projects (
    id SERIAL PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    code VARCHAR(50) UNIQUE NOT NULL,
    description TEXT,
    start_date DATE,
    end_date DATE,
    budget DECIMAL(15,2),
    status VARCHAR(50) DEFAULT 'Planning',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Phases Table
CREATE TABLE phases (
    id SERIAL PRIMARY KEY,
    project_id INTEGER REFERENCES projects(id),
    name VARCHAR(255) NOT NULL,
    sequence INTEGER NOT NULL,
    description TEXT,
    status VARCHAR(50) DEFAULT 'Not Started',
    start_date DATE,
    end_date DATE,
    gate_criteria JSONB,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Roles Table
CREATE TABLE roles (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    description TEXT,
    responsibilities TEXT[],
    tools TEXT[],
    skills_required TEXT[]
);

-- Team Members Table
CREATE TABLE team_members (
    id SERIAL PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    email VARCHAR(255) UNIQUE NOT NULL,
    role_id INTEGER REFERENCES roles(id),
    department VARCHAR(100),
    availability DECIMAL(3,2) DEFAULT 1.0,
    skills TEXT[],
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Tasks Table
CREATE TABLE tasks (
```

```
    id SERIAL PRIMARY KEY,
    phase_id INTEGER REFERENCES phases(id),
    name VARCHAR(255) NOT NULL,
    description TEXT,
    assigned_to INTEGER REFERENCES team_members(id),
    status VARCHAR(50) DEFAULT 'To Do',
    priority VARCHAR(20) DEFAULT 'Medium',
    due_date DATE,
    estimated_hours DECIMAL(6,2),
    actual_hours DECIMAL(6,2),
    dependencies INTEGER[],
    sequence INTEGER,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Deliverables Table
CREATE TABLE deliverables (
    id SERIAL PRIMARY KEY,
    task_id INTEGER REFERENCES tasks(id),
    name VARCHAR(255) NOT NULL,
    description TEXT,
    status VARCHAR(50) DEFAULT 'Pending',
    due_date DATE,
    acceptance_criteria TEXT[],
    file_path VARCHAR(500),
    approved_by INTEGER REFERENCES team_members(id),
    approved_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Challenges Table
CREATE TABLE challenges (
    id SERIAL PRIMARY KEY,
    project_id INTEGER REFERENCES projects(id),
    description TEXT NOT NULL,
    category VARCHAR(100),
    severity VARCHAR(20),
    impact TEXT,
    solution TEXT,
    strategy TEXT,
    status VARCHAR(50) DEFAULT 'Open',
    owner_id INTEGER REFERENCES team_members(id),
    identified_date DATE DEFAULT CURRENT_DATE,
    resolved_date DATE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Hardware Items Table
CREATE TABLE hardware_items (
    id SERIAL PRIMARY KEY,
    project_id INTEGER REFERENCES projects(id),
    category VARCHAR(100),
    name VARCHAR(255) NOT NULL,
```

```
    model VARCHAR(255),
    specifications JSONB,
    quantity INTEGER,
    unit_cost DECIMAL(12,2),
    total_cost DECIMAL(15,2),
    vendor VARCHAR(255),
    lead_time_days INTEGER,
    status VARCHAR(50) DEFAULT 'Pending',
    order_date DATE,
    delivery_date DATE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Tools Table
CREATE TABLE tools (
    id SERIAL PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    category VARCHAR(100),
    license_type VARCHAR(50),
    annual_cost DECIMAL(12,2),
    documentation_url VARCHAR(500),
    roles INTEGER[],
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- KPIs Table
CREATE TABLE kpis (
    id SERIAL PRIMARY KEY,
    project_id INTEGER REFERENCES projects(id),
    name VARCHAR(255) NOT NULL,
    baseline_value DECIMAL(12,4),
    target_value DECIMAL(12,4),
    current_value DECIMAL(12,4),
    unit VARCHAR(50),
    measurement_method TEXT,
    frequency VARCHAR(50),
    last_measured TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Weekly Status Table
CREATE TABLE weekly_status (
    id SERIAL PRIMARY KEY,
    project_id INTEGER REFERENCES projects(id),
    week_ending DATE,
    overall_status VARCHAR(20),
    accomplishments TEXT[],
    next_week_plans TEXT[],
    risks TEXT[],
    blockers TEXT[],
    budget_spent DECIMAL(15,2),
    schedule_variance INTEGER,
    created_by INTEGER REFERENCES team_members(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
```

```
);

-- Indexes
CREATE INDEX idx_tasks_phase ON tasks(phase_id);
CREATE INDEX idx_tasks_assigned ON tasks(assigned_to);
CREATE INDEX idx_tasks_status ON tasks(status);
CREATE INDEX idx_deliverables_task ON deliverables(task_id);
CREATE INDEX idx_challenges_project ON challenges(project_id);
CREATE INDEX idx_hardware_project ON hardware_items(project_id);
CREATE INDEX idx_kpis_project ON kpis(project_id);
```

7.3 Sample Data Insert

```
-- Insert Project
INSERT INTO projects (name, code, description, start_date, end_date, budget, status)
VALUES (
    'AI-Based Monitoring Solution for Oil & Gas',
    'OG-AI-MON-2025',
    'ML/DL/CV/RAG solution for predictive maintenance and safety monitoring',
    '2025-01-01',
    '2026-06-30',
    5980000.00,
    'In Progress'
);

-- Insert Phases
INSERT INTO phases (project_id, name, sequence, status, start_date, end_date) VALUES
(1, 'Discovery & Requirements', 1, 'Completed', '2025-01-01', '2025-02-28'),
(1, 'Architecture & Design', 2, 'Completed', '2025-03-01', '2025-04-30'),
(1, 'Development', 3, 'In Progress', '2025-05-01', '2025-10-31'),
(1, 'Pilot Deployment', 4, 'Not Started', '2025-11-01', '2026-01-31'),
(1, 'Full Rollout', 5, 'Not Started', '2026-02-01', '2026-05-31'),
(1, 'Operations', 6, 'Not Started', '2026-06-01', NULL);

-- Insert Roles
INSERT INTO roles (name, description, responsibilities, tools, skills_required) VALUES
('Project Manager', 'Overall project delivery',
 ARRAY['Planning', 'Budget', 'Risk', 'Stakeholder'],
 ARRAY['Jira', 'MS Project', 'Confluence'],
 ARRAY['PMP', 'Agile', 'Oil & Gas experience']),
('ML Engineer', 'ML/DL model development',
 ARRAY['Model training', 'Optimization', 'Deployment'],
 ARRAY['PyTorch', 'TensorFlow', 'MLflow'],
 ARRAY['Python', 'Deep Learning', 'MLOps']),
('Hardware Engineer', 'Sensor and edge hardware',
 ARRAY['Sensor selection', 'PCB design', 'Installation'],
 ARRAY['KiCad', 'AutoCAD', 'Oscilloscope'],
 ARRAY['Electronics', 'Industrial sensors', 'ATEX']);
```

PART 8: BLOG SERIES TEMPLATE

Blog 1: BRD and Problem Statement (This Document)

Content: Full BRD as documented above

Blog 2: Architecture Deep-Dive

Topics: Edge architecture, cloud platform, data flows, security

Blog 3: Hardware Selection for Oil & Gas

Topics: Sensors, edge compute, cameras, ATEX compliance

Blog 4: Data Pipeline Development

Topics: Kafka, SCADA integration, time-series storage

Blog 5: ML Model - Predictive Maintenance

Topics: Feature engineering, model training, edge deployment

Blog 6: DL Model - Anomaly Detection

Topics: Autoencoder design, threshold tuning, real-time inference

Blog 7: CV Model - Corrosion Detection

Topics: Dataset creation, YOLO training, POD validation

Blog 8: CV Model - PPE Compliance

Topics: Multi-class detection, video pipeline, privacy

Blog 9: CV Model - Gas Leak Detection

Topics: Thermal imaging, multi-modal fusion, alert logic

Blog 10: RAG System - Knowledge Assistant

Topics: Document ingestion, embeddings, LLM fine-tuning, chatbot

Blog 11: Dashboard and Visualization

Topics: Control room UI, mobile app, Grafana

Blog 12: Integration - SAP PM and Historian

Topics: Work order automation, data synchronization

Blog 13: Testing and Validation

Topics: Unit tests, integration tests, UAT, performance

Blog 14: Pilot Deployment

Topics: Installation, training, shadow mode, go-live

Blog 15: Operations and Continuous Improvement

Topics: Monitoring, retraining, feedback loops

APPENDIX A: GLOSSARY

Term	Definition
ATEX	European directive for equipment in explosive atmospheres
CV	Computer Vision
DCS	Distributed Control System
DL	Deep Learning
ESD	Emergency Shutdown
FMEA	Failure Mode and Effects Analysis
HAZOP	Hazard and Operability Study
IEC 62443	Industrial cybersecurity standard
LDAR	Leak Detection and Repair
ML	Machine Learning
MOC	Management of Change
MTTD	Mean Time to Detection
MTTR	Mean Time to Repair
OPC-UA	Open Platform Communications Unified Architecture
POD	Probability of Detection
PPE	Personal Protective Equipment
PSM	Process Safety Management
RAG	Retrieval-Augmented Generation
RBI	Risk-Based Inspection
SCADA	Supervisory Control and Data Acquisition
TRIR	Total Recordable Incident Rate

APPENDIX B: APPROVAL SIGNATURES

Role	Name	Signature	Date
Executive Sponsor			
Business Owner			
IT Director			
HSE Director			
Project Manager			

Document Control

Version	Date	Author	Changes
1.0	Dec 2025	AI Team	Initial draft

END OF DOCUMENT